

FJÄLL AFFORDABLE DEVELOPMENT

Impact-Driven Mass Housing Solutions & Advanced Structural Manufacturing.
Solusi Perumahan Massal Berdampak Tinggi & Manufaktur Struktural Lanjutan.



About Fjäll Group



Established to combat the global housing backlog, Fjäll Group through its product Fjäll Affordable Development 'FAD' is an advanced manufacturing company specialising in composite panel infrastructure.

Didirikan untuk mengatasi backlog perumahan global, Fjäll Group melalui produk Fjäll Affordable Development 'FAD' adalah perusahaan manufaktur canggih yang berspesialisasi dalam infrastruktur panel komposit.

Traditional concrete is slow, resource-heavy, and highly vulnerable to seismic risks. Through perpetual research and development, we provide rapid, scalable, and earthquake-resistant solutions for mass housing development, redefining affordability and safety.

Beton tradisional lambat dan rentan terhadap risiko seismik. Melalui penelitian dan pengembangan, kami memberikan solusi perumahan massal yang cepat, terukur, dan tahan gempa.

VISION

"Resilient Infrastructure Worldwide"

Infrastruktur Tangguh di Seluruh Dunia

MISSION

"Safe, Fast, and Affordable Housing in Every District"

Perumahan yang Aman, Cepat, dan Terjangkau di Setiap Daerah

The Fjäll Advantage



Hearing what the market demands is the key to designing the right innovation. Our technology replaces traditional masonry with pre-engineered perfection.

Mendengar tuntutan pasar adalah kunci untuk merancang inovasi. Teknologi kami menggantikan metode tradisional dengan presisi pracetak.



Ultra-Fast Deploy Build time of just 1-3 hours per unit, unaffected by rainy weather.



Disaster Resilient Monolithic structure absorbs shockwaves up to 9.0 Magnitude.



Capital Efficiency Reduces reliance on heavy machinery and on-site labor drastically.



Eco-Engineered 100% VOC free, mold-proof, rust-proof, termite-proof material.

Engineering Overview



Fjäll Group maximises production performance with precision technology. Our composite panels consist of a High-Density EPS core bound by a 200-400g/m² fiberglass matrix and proprietary IQ Structural Resin.

Kami memaksimalkan kinerja produksi dengan teknologi presisi. Panel kami terdiri dari inti EPS berkepadatan tinggi yang diikat matriks fiberglass dan Resin Struktural IQ.



36m²

Base Unit Footprint

Luas Unit Dasar

375kg

Structural Weight

Total Berat Struktural

10

Primary Panels Req.

Kebutuhan Panel Utama

100+

Years Est. Lifespan

Estimasi Umur Bangunan

Standard 36m² Specifications



FAD Panel System

Constructed using exactly 10 high-strength FAD composite panels (6m x 2.4m each).

Dibangun menggunakan tepat 10 panel komposit FAD berkekuatan tinggi.

Includes standard layout with 3 UPVC main windows and 1 upper ventilation window (boven).

Termasuk tata letak standar dengan 3 jendela utama UPVC dan 1 jendela atas.

Architecturally flexible system allowing for vertical expansion up to 2 floors if required.

Sistem fleksibel yang memungkinkan perluasan vertikal hingga 2 lantai.

Design Variants

Available in four professional standard exterior finishes to match urban planning needs:

Tersedia dalam empat warna eksterior standar profesional:



Olive Green



Earthy Brown



Golden Yellow



Royal Purple



Standard 36m² Layout



Core Structure	Materials
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Wall	EPS/Fiberglass/Plamir (6-30cm)
Foundation	Stone/Block Base (50cm)
Fixture	Sandwich Panel, UPVC
Roofing	EPS/Go-Green/Spandex

Zone	Dimensions / Allocation
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Master Bedroom	3.0m x 3.0m (9.0 m ²)
Secondary Room	3.0m x 3.0m (9.0 m ²)
Living Area	3.0m x 4.0m (12.0 m ²)
Wet Core (Bath/Kit)	Integrated 6.0 m ² block
Foundation Req.	50cm Stone/Block Perimeter

1 unit requires exactly **10 FAD panels** (6 × 2.4m).
1 unit dibangun menggunakan 10 panel FAD.

Total Weight
375Kg

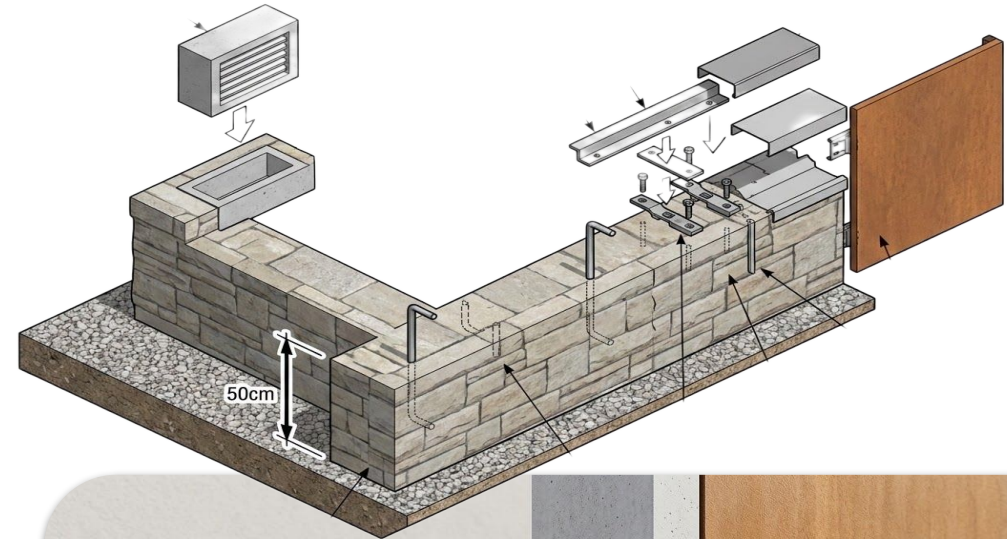


Lightweight Foundation

The entire 36m² structure weighs only 375kg, completely eliminating the need for heavy, expensive traditional foundations.

Berat seluruh struktur 36m² hanya 375kg, menghilangkan kebutuhan pondasi tradisional yang berat dan mahal.

We utilise a simple 50cm stone/block base with high-strength 6mm steel anchors embedded every 1 meter for maximum grip. Kami menggunakan dasar batu 50cm dengan jangkar baja 6mm berkekuatan tinggi yang ditanam setiap 1 meter.

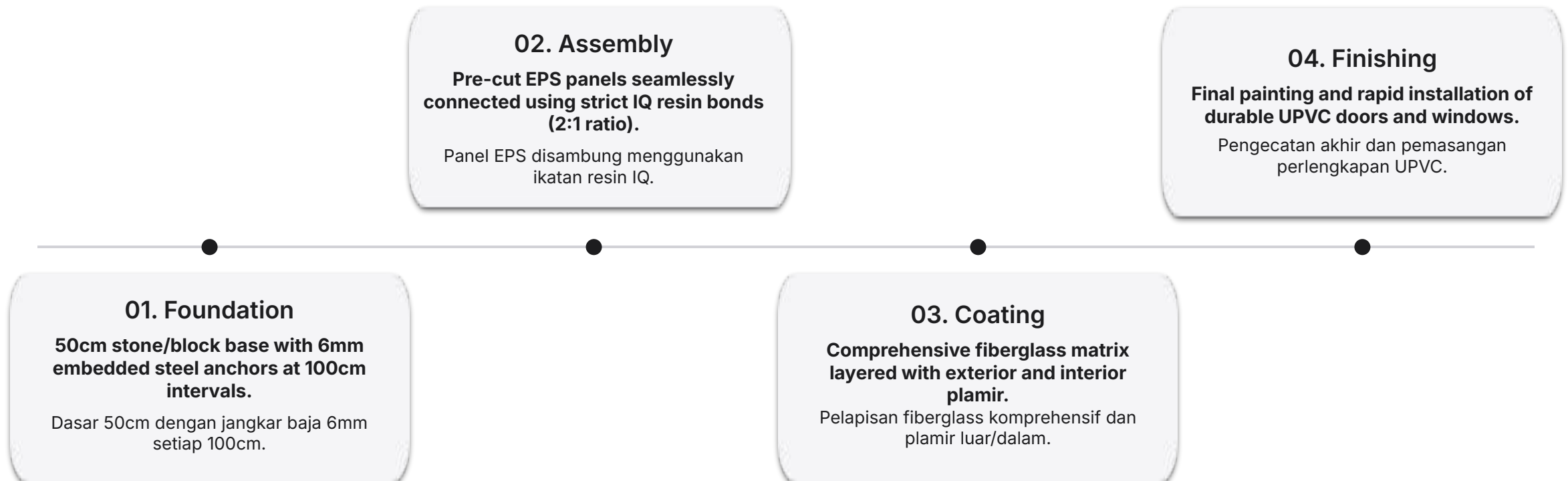


Construction Protocol



Precision engineering enables a complete superstructure build time of just 1 to 3 hours per unit, unaffected by rainy weather conditions.

Waktu bangun struktur atas hanya 1-3 jam per unit, tidak terpengaruh oleh cuaca hujan.



Global Compliance & Standards



Fire Resistance Rating (REI 60): Certified to withstand extreme direct fire exposure for over 30 minutes without structural failure.

Ketahanan Api: Bersertifikat menahan paparan api selama 30+ menit.



Toxic Gas & TVOC Verification: Passed stringent lab evaluations. 100% VOC free materials guaranteeing zero hazardous gas emissions.

Verifikasi Gas Beracun: 100% bebas VOC, nol emisi gas berbahaya.



Structural Integrity Tests: Verified extremely high compression load tolerance (48,278 N/m) and 4-point bending strength (3,569 N/m²).

Integritas Struktural: Kompresi sangat tinggi (48.278 N/m).



Seismic Resistance: Monolithic composite structure dynamically absorbs shockwaves, withstanding earthquakes up to 9.0 Magnitude.

Ketahanan Gempa: Struktur komposit menahan guncangan hingga 9.0 SR.

Versatile Deployment



Mass Residential
Perumahan Massal / Tapak



Educational Facilities
Fasilitas Sekolah / Kampus



Mass Dormitories
Asrama Kepadatan Tinggi

Commercial Proposition



Construction Component / Item	Included	Developer Scope
Stone Foundation (50cm) & Septic Tank Pondasi Batu (50cm) & Tangki Septic	✓	
FAD Composite Panel Walls & Roofing Sistem Dinding & Atap Komposit FAD	✓	
UPVC Doors, Windows & Exterior Coating Pintu, Jendela UPVC & Cat Eksterior	✓	
Electrical Installation & Interior Finishing Instalasi Listrik & Cat Interior		—

BASELINE TURNKEY VALUE (TYPE 36M²)

Rp 50.000.000,-

ADDITIONAL AREA SCALING

Rp 1.400.000 / m²



Resilient Infrastructure
Worldwide



fjallsolutions.com



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Testing Result Report



Product Information

SkinTec EP



Product Description

SkinTec EP is a FRP laminate, which is produced in a continuous process to guarantee a consistent high quality. Due to the high glass-fibre content and the chosen glass textiles extremely high mechanical performance can be achieved. A specially developed resin enables easy bonding and painting of the laminates.

Product Range

Laminates are produced in thicknesses between 0.8 mm and 3 mm and are showing an unfinished surface. Typically one side is coated white. Tailor-made combinations of glass textiles and stitched mats are available on request, too.

Sizes

Length: endless, as coils
Width: up to 3000 mm

Processing

Machining of SkinTec EP is very easy. Use tools for wood machining for cutting, sawing, machining, drilling, grinding and bending. Suitable joining techniques for the laminates are e.g. gluing and nailing.

Environmental friendly

The production and therefore the laminates are absolutely free of VOCs. There is no solvent (e.g. such as styrene) used during the entire production process and the laminates do not exhibit an odour or smell.

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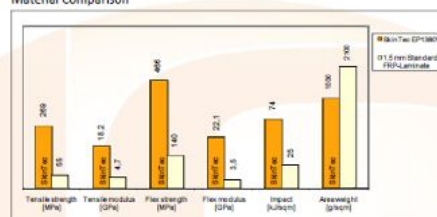
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Mechanical & Physical Properties¹

Property	Standard ASTM	Unit	SkinTec EP1200WE	SkinTec EP1320WE	SkinTec EP1320WE	SkinTec EP1480WE
Thickness		mm	0.43	0.9	1.4	1.9
Weight		g/m ²	250	1550	2300	2650
Fibre content	D2554	%	62	60	62	60
Tensile strength	D3039	MPa	294	269	215	209
Tensile modulus	D3039	GPa	13.9	13.2	13.0	13.8
Flex strength	D790	MPa	468	486		379
Flex modulus	D790	GPa	22.3	22.1		12.3
Impact strength, Izod	D2560	kJ/m ²	36	74		93
Barcol hardness	D2583	Barcol	59	59	59	59
Tg	D3418	°C	96	96	96	96
Coefficient of thermal expansion	D696, D331, D3015/29	10 ⁻⁴ K ⁻¹	15.4	15.4	15.4	15.4
Thermal conductivity	E1530	W/mK	0.23	0.23	0.23	0.23
Moisture absorption (24 hr soak)	D570-98	%	0.13	0.13	0.13	0.13

Material Comparison



Disclaimer

The mechanical properties in this data sheet represent our current state of technical knowledge. The properties are based on testing and calculations. As variations are possible in production, we cannot guarantee particular properties or reliability for any specific application and the data should not be used to establish specification limits nor used alone as basis of design. This information may be changed as new knowledge and experience becomes available. IQ Tec Germany GmbH gives no warranties and assumes no liability in connection with any use of this information.

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REPORT

Date: 2012-12-20 | Reference: FX18474-1 | Page: 4 (5)

4 Classification and field of application

4.1 Reference of classification

This classification has been carried out in accordance with Clause 7 of EN 13501-2:2007+A1:2009.

4.2 Classification

This element has been classified:

Fire resistance classification: REI 60

4.3 Field of application

This classification is valid for the following end use applications:

Direction:	Both directions
Height:	Decrease, ≤ 3 meter
Thickness:	Increase, ≥ 278 mm
Thickness of component materials:	Increase
Linear dimensions of boards or dimensions of panels, but not thickness:	Decrease
Number of horizontal joints:	Increase, none or more
Applied load:	Decrease, ≤ 6.1 kN/meter wall
Width:	Increase, ≥ 3 meter

SP Technical Research Institute of Sweden



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Classified by: **SP Technical Research Institute of Sweden**
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CLASSIFICATION OF FIRE RESISTANCE IN ACCORDANCE WITH EN 13501-2:2007+A1:2009

Specimen: KJ Tec Germany GmbH
Name: Wall element Str. 24
DE-73236 KIRCHHEIM/TECK
Tyskland

Prepared by: SP Technical Research Institute of Sweden
Max 057
SE-831 15
Svante

Product name: None

Classification report No.: FX18474-1

Date of issue: December 20, 2012

This classification report consists of 5 pages.

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